A Project Report on

Wind Energy Generation Models: Physical, Statistical & Hybrid

Submitted in partial fulfilment of the requirements for the award of the degree of

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In

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Submitted by

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CERTIFICATE

This is to certify that the Project work entitled "**Wind Energy Generation Models: Physical, Statistical & Hybrid**" is a bonafide record of work carried out by **Karthikeya Chanda (21EEB0B18), Ishaan Yadav (21EEB0B24)** of Final year B.Tech (Electrical and Electronics Engineering) , in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in "Electrical and Electronics Engineering" at National Institute of Technology, Warangal during the academic year 2021-2025.

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

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ABSTRACT

As global efforts intensify to transition toward sustainable and carbon-free energy systems, wind energy has emerged as a prominent and promising renewable energy source due to its scalability, widespread availability, and zero emissions during operation. However, the intrinsic variability and intermittency of wind pose significant challenges in forecasting energy output and ensuring grid stability. Reliable and accurate wind power prediction has thus become essential for planning, operation, and integration of wind energy systems into the existing electricity infrastructure. This project report presents a comprehensive exploration and comparative analysis of various wind energy generation models, encompassing physical, statistical, machine learning, deep learning, and hybrid modeling techniques. The study aims to identify the most effective predictive approaches for estimating wind power output under diverse meteorological and geographical conditions.

The modeling process is structured across three distinct categories: physical models, statistical models, and hybrid or deep learning-based models. Physical models, such as the Actuator Disk Model (ADM) and the Blade Element Momentum (BEM) model, are grounded in aerodynamic theory and fluid dynamics. These models rely on assumptions of incompressible and steady airflow to mathematically capture the energy conversion process in wind turbines. While ADM provides a theoretical efficiency ceiling—represented by the Betz Limit—BEM extends this foundation by incorporating blade-specific characteristics and resolving aerodynamic forces at discrete blade segments, leading to more detailed estimations of power, thrust, and torque.

Statistical models are data-driven approaches that leverage historical time-series data to uncover autocorrelations and trends in wind behavior. The project explores classical models such as Auto-Regressive (AR), Moving Average (MA), and Auto-Regressive Moving Average (ARMA). These models offer computational efficiency and interpretability but are limited in capturing the non-linear and multi-variable nature of wind energy systems. Regression-based models such as Linear Regression, Multiple Linear Regression, and Polynomial Regression were also implemented to map meteorological variables like wind speed, temperature, and humidity to turbine power output. Among these, Polynomial Regression demonstrated better performance by accommodating non-linearity in the data.

To address the limitations of classical approaches, the project integrated machine learning and deep learning models. The K-Nearest Neighbors (KNN) model, known for its non-parametric behavior, provided superior accuracy among classical models, with minimal assumptions about data distribution. Random Forest, an ensemble method based on decision trees, demonstrated robustness to noise and delivered balanced performance across various metrics.

For capturing more complex, temporal dependencies in the dataset, several deep learning models were developed. The Artificial Neural Network (ANN) showed exceptional accuracy by learning intricate patterns within the data. Recurrent Neural Networks (RNN) and their variant, Gated Recurrent Units (GRU), were implemented to leverage sequential correlations in time-series data. These architectures proved particularly effective in modeling the temporal fluctuations characteristic of wind speed and turbine output. Additionally, Convolutional Neural Networks (CNNs) were adapted to extract features from multivariate input sequences and demonstrated competitive performance in short-term forecasting tasks.

A key highlight of this project is the implementation of a hybrid model: the Adaptive Neuro-Fuzzy Inference System (ANFIS). ANFIS merges the transparency of fuzzy logic with the adaptive capabilities of neural networks, offering both interpretability and reasonable accuracy. This makes it especially suitable for applications where model transparency and expert rule-based reasoning are critical, such as operational decision-making and regulatory compliance.

All models were trained and evaluated using a rich dataset comprising wind characteristics (speed, direction, gusts) and meteorological parameters (temperature, humidity, dew point), collected at multiple heights. Preprocessing steps included data cleaning, normalization, feature selection, and chronological splitting to preserve temporal integrity. Model performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.

The findings indicate that while classical statistical models offer a foundation for understanding wind dynamics, modern deep learning models—particularly ANN and RNN—demonstrated superior predictive power. KNN and ANFIS also stood out as robust alternatives offering high accuracy and interpretability, respectively. Overall, this report concludes that hybrid and deep learning models provide the most promising avenue for accurate and scalable wind power forecasting and should be prioritized in future smart grid and renewable energy integration efforts.

CHAPTER 1
INTRODUCTION

1.1 Background

The transition to renewable energy sources is a global imperative in light of climate change, energy security concerns, and the depletion of fossil fuel resources. Among various renewable energy technologies, wind energy has emerged as one of the most promising due to its abundance, sustainability, and zero carbon emissions during operation. Wind power systems are being increasingly integrated into national grids worldwide, leading to a need for accurate modeling and forecasting to ensure reliability and efficiency.

However, wind energy presents significant challenges in prediction and integration due to the inherent variability and intermittency of wind. These characteristics introduce complex, non-linear behaviors into power generation that cannot be accurately captured using simple linear models. Hence, there is a pressing need for advanced models that can analyze wind data and produce reliable power forecasts under different atmospheric and geographic conditions.

1.2 Project Objective

The primary objective of this project is to develop, implement, and compare various models to predict the power output of wind turbines based on meteorological and wind parameters. The models range from physically-based analytical formulations to data-driven methods using statistical techniques, machine learning (ML), and deep learning (DL). A hybrid approach is also explored to capture the strengths of both traditional and modern methods.

This comprehensive modeling aims to:

- Improve forecasting accuracy for wind energy output.
- Facilitate better planning and operation of wind farms.
- Provide insights into model suitability for real-world deployment across varying conditions.

1.3 Scope of the Project

The project encompasses the end-to-end pipeline of wind power prediction, starting from data acquisition and preprocessing to model training, evaluation, and comparison. The scope includes:

- Development of physical models based on aerodynamic principles (e.g., Actuator Disk Model and BEM).
- Construction of statistical models for time-series forecasting (e.g., AR, MA, ARMA).
- Implementation of machine learning models such as Linear Regression, Polynomial Regression, and KNN.
- Exploration of deep learning architectures including ANN, RNN, GRU, and CNN for complex temporal 14odelling.
- Integration of models into a hybrid framework like ANFIS to handle uncertainties effectively.

Each model is evaluated using standard metrics to determine their efficacy and limitations, with the goal of identifying the most robust method for accurate and scalable wind energy forecasting.

1.4 Tools & Technologies

The project was implemented using the Python programming language due to its robust ecosystem of data science and machine learning libraries. The key tools and technologies include:

- NumPy and pandas for data manipulation and preprocessing.
- scikit-learn for statistical and ML model implementation.
- TensorFlow and Keras for building and training deep learning architectures.
- matplotlib and seaborn for visualization and performance analysis.
- Kaggle API to fetch meteorological datasets containing features such as wind speed, temperature, and power output.

1.5 Organization of the Report

The report is structured to reflect a logical progression through the project:
- Chapter 1: Introduction – Provides background, objectives, and the scope of the project.
- Chapter 2: Literature Review – Summarizes existing research and forecasting techniques in wind energy.

- Chapter 3: Methodology – Describes the models, dataset, and evaluation strategies used in the study.
- Chapter 4: Results and Discussion – Presents model outputs, error comparisons, and result interpretations.
- Chapter 5: Conclusion and Future Work – Concludes the study and outlines directions for further research.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Wind energy forecasting plays a critical role in integrating renewable energy into the power grid. Due to the intermittent and non-linear nature of wind, predicting power output from wind turbines poses significant challenges. Over the years, a wide range of models has been proposed to forecast wind power, ranging from physical models based on fluid dynamics to data-driven statistical and artificial intelligence (AI) approaches.

This chapter provides a detailed survey of the literature and theoretical foundations of the models employed in this project. These models are categorized as:

- **Physical Models**
- **Statistical Models**
- **Machine Learning Models**
- **Deep Learning Models**
- **Hybrid Models**

Forecasting wind energy generation has been an active area of research due to its significance in planning, grid integration, and reliability assessment. A range of models have been proposed, from classical statistical methods to sophisticated AI-driven approaches. This chapter summarizes notable literature relevant to each category.

2.1.1 Reliability Evaluation Using Wind Models

A foundational study by Rajesh Karki et al. focused on the development of simplified wind power generation models for reliability evaluation. The study utilized historical wind speed data from three Canadian sites (Swift Current, North Battleford, and Toronto) to validate the model's

capability. The stochastic nature of wind was addressed using Monte Carlo simulations, which allowed the researchers to estimate system performance metrics such as Loss of Load Probability (LOLP).

2.2 Forecasting Techniques Using ANN and ARIMA

David Barbosa et al. examined multiple forecasting techniques for ultra-short to long-term wind speed prediction. Their approach combined Artificial Neural Networks (ANNs) and Autoregressive Integrated Moving Average (ARIMA) models. The dataset was sourced from the Petrolina station in Brazil and spanned over 13 years. The ANN model was effective in capturing non-linear dependencies, while ARIMA provided a solid baseline for statistical forecasting.

2.3 Hybrid Approaches and Time-Series Models

Recent studies have moved toward hybrid approaches to combine the strengths of individual methods. For instance, ANFIS models merge fuzzy logic with neural networks to handle data uncertainty and imprecision effectively. These models have demonstrated superior performance in real-time applications.

2.4 Summary of Reviewed Literature

Study	**Model**	**Dataset**	**Techniques**	**Outcome**
Rajesh Karki et al.	Wind Speed Model	Canada (3 sites)	Monte Carlo Simulation	Reliability Evaluation
David Barbosa et al.	ANN & ARIMA	Petrolina, Brazil	Time-series Forecasting	Short/Long-term Accuracy

Various Recent Works	ANFIS, Hybrid Models	Multisite Datasets	Hybrid Learning, Fuzzy Logic	Enhanced Robustness

Table 2.1

CHAPTER 3
METHODOLOGY

3.1 Data Acquisition and Description

The foundation of any data-driven wind energy forecasting model lies in the quality and granularity of the data used. For this study, a comprehensive dataset was obtained, comprising both meteorological parameters and actual wind power generation measurements. This dataset serves as the cornerstone for building physical, statistical, and hybrid models to evaluate wind energy potential and prediction accuracy.

The dataset includes time-series data with a high temporal resolution, typically sampled at regular intervals (e.g., hourly). Each observation corresponds to a timestamp and consists of various meteorological features alongside the actual power output from a wind turbine or wind farm. These features can be grouped into three main categories:

1. **Wind Characteristics**:
    - **Windspeed_10m** and **Windspeed_100m**: These represent the wind speed measured at 10 meters and 100 meters above ground level, respectively. Wind speed is the most critical variable influencing power generation, as the power output of a wind turbine is approximately proportional to the cube of the wind speed.
    - **Winddirection_10m** and **Winddirection_100m**: These provide the direction from which the wind is blowing at the corresponding heights. Wind direction is essential for turbine yaw control and for capturing wind flow alignment with the rotor.
    - **Windgusts_10m**: Represents short-duration bursts of high-speed wind. Gusts can affect turbine load and influence power spikes.
2. **Meteorological Variables**:
    - **Temperature_2m**: The ambient temperature at 2 meters height. Temperature affects air density, which in turn influences the kinetic energy available in the wind.

- **Relativehumidity_2m**: Humidity levels can affect atmospheric pressure and density, indirectly influencing wind power generation.
- **Dewpoint_2m**: Used in conjunction with temperature to understand moisture content and predict fog or icing, which may reduce turbine efficiency.

3. **Target Variable**:
   - **Power**: This is the key output variable, representing the electrical power generated by the wind turbine (in kilowatts or megawatts, depending on the dataset). This value is modeled or predicted using the above input features.

This diverse set of features provides a rich ground for exploring various modeling approaches. While physical models primarily utilize wind speed and turbine-specific parameters, statistical and deep learning models make use of a broader set of predictors to improve accuracy.

3.2 Data Preprocessing

Raw data collected from wind turbines and meteorological sources often contain inconsistencies, missing values, or noise. These issues can negatively impact the accuracy of both statistical and machine learning models. Therefore, rigorous data preprocessing is crucial to ensure the reliability and effectiveness of the predictive models. The preprocessing pipeline implemented in this project involved several key steps, described in detail below.

3.2.1 Handling Missing and Anomalous Data

Wind datasets often suffer from sensor malfunctions, communication errors, or adverse weather conditions that result in missing or corrupted data. Observations with missing values were either:

- **Removed**: If the proportion of missing data was small and occurred randomly.
- **Interpolated**: Using forward fill or linear interpolation techniques to estimate missing values based on surrounding data points.

Additionally, outliers and anomalous readings (e.g., negative wind speeds or unrealistically high gusts) were identified using statistical thresholds (such as z-scores or interquartile ranges) and either corrected or excluded from the analysis.

3.2.2 Feature Scaling

Many models—especially distance-based models like KNN and gradient-based models like ANN and RNN—require input features to be scaled to a consistent range. Standardization was applied to the dataset using the **z-score method**:

$$x' = \frac{x - \mu}{\sigma}$$

Where x is the original value, μ is the mean, and σ is the standard deviation. This ensures that all features contribute equally to the model's learning process, preventing bias toward variables with larger numeric ranges.

3.2.3 Feature Selection and Engineering

Only relevant features that showed high correlation with the power output were selected for model input. This reduces the dimensionality of the problem and eliminates noise from irrelevant data. In some cases, **derived features** (e.g., wind speed cubed, direction cosine components) were computed to better capture non-linear relationships in the data.

3.2.4 Temporal Structuring

Since wind data is inherently time-series in nature, appropriate temporal ordering was preserved. For models like AR, MA, ARMA, and RNN that rely on sequential patterns, lag variables and time-windowed datasets were created to provide historical context to the models.

3.2.5 Dataset Splitting

Finally, the processed dataset was split into:

- **Training Set**: Used to fit the model (typically 70–80% of the data),
- **Test Set**: Used to evaluate the model's generalization ability on unseen data (typically 20–30%).

In time-series forecasting, **chronological splitting** (rather than random splitting) was employed to avoid data leakage from future observations into the training set.

3.3 Physical Models in Wind Energy

Physical models in wind energy generation aim to describe the conversion of wind's kinetic energy into mechanical or electrical energy using established physical laws. These models are not reliant on historical data like statistical or machine learning models. Instead, they are rooted in classical mechanics, particularly in the fields of aerodynamics and fluid dynamics. Among these, the **Actuator Disk Model (ADM)** stands out as a foundational approach. It simplifies the real wind turbine to an idealized, non-rotating disk that extracts energy from the wind uniformly across its surface. Though simplistic, this model lays the groundwork for understanding more complex turbine interactions.

3.3.1 The Actuator Disk Model (ADM)

The Actuator Disk Model treats the rotor of a wind turbine as a thin, porous, non-rotating disk. As wind passes through this disk, it slows down due to energy extraction. The model is derived using **conservation of mass**, **linear momentum**, and **Bernoulli's principle**. It assumes an ideal flow that is incompressible, steady, and non-turbulent, with no rotational wake effects.

In the ADM framework, let the freestream wind velocity (before encountering the turbine) be V_∞, and the wind velocity at the disk be V_d. Downstream of the disk, the wind slows further to V_w. The **induction factor** a is defined as the fractional reduction in wind speed at the disk compared to the freestream:

$$V_d = V_\infty(1 - a)$$

The wind speed in the far wake is then:

$$V_w = V_\infty(1 - 2a)$$

Power Extracted by the Turbine

The power extracted from the wind by the turbine is the rate of change of kinetic energy of the air passing through the rotor. Using the conservation of energy and assuming a uniform velocity across the rotor disk, the power P can be expressed as:

$$P = \frac{1}{2}\rho A V_\infty^3 \cdot 4a(1 - a)^2$$

Here:

- ρ is the air density (typically 1.225 kg/m³ at sea level),
- A is the swept area of the rotor ($A = \pi R^2$, where R is the blade radius),
- V_∞ is the wind velocity before encountering the rotor,
- a is the axial induction factor.

The term $4a(1-a)^2$ determines the efficiency of energy extraction. The maximum value of this term occurs when $a = \frac{1}{3}$. Substituting this into the power equation gives the **maximum extractable power**, known as the **Betz limit**:

$$P_{max} = \frac{16}{27} \cdot \frac{1}{2} \rho A V_\infty^3$$

This implies the maximum efficiency or **power coefficient** C_p is:

$$C_{p_{max}} = \frac{P_{max}}{\frac{1}{2} \rho A V_\infty^3} = \frac{16}{27} \approx 0.593$$

Therefore, no wind turbine based solely on momentum theory can capture more than 59.3% of the kinetic energy from the wind, regardless of design.

Significance and Applications

The Actuator Disk Model serves as a fundamental theoretical tool in wind turbine analysis and preliminary design. It allows engineers to estimate how much power a turbine can potentially extract under idealized conditions. While the assumptions made—such as uniform wind speed and no rotational effects—do not hold true in practical scenarios, ADM offers a benchmark against which real turbine performance can be measured.

Moreover, this model forms the basis for more detailed physical models like the **Blade Element Momentum Theory (BEMT)**, which adds rotational effects and considers the geometry of individual blades. Understanding ADM is thus crucial for progressing to more sophisticated aerodynamic modeling of turbines.

Limitations of ADM

Despite its elegance, the ADM is inherently limited by its assumptions. It cannot account for blade geometry, tip losses, wake rotation, or turbulent inflows. These factors significantly influence real-world turbine performance. As a result, while the ADM provides valuable insight into the theoretical upper bounds of energy conversion, practical turbine design and analysis require more complex methods, often combined with empirical data or computational simulations.

3.3.2 Blade Element Momentum (BEM) Model

The **Blade Element Momentum (BEM) theory** is a refined physical model used extensively in the analysis and design of wind turbines. It combines the principles of momentum theory—used in the Actuator Disk Model—with the blade element theory that accounts for the aerodynamic forces acting on individual blade segments. By integrating both momentum conservation and detailed blade geometry, the BEM model provides a much more realistic and practical framework for predicting the performance of horizontal-axis wind turbines (HAWTs). It is especially useful for estimating the power output, thrust, and torque of a wind turbine across different operating conditions.

The Core Idea Behind BEM

In the BEM method, the turbine blade is divided into several small elements along its length (radially from hub to tip). For each of these **blade elements**, the aerodynamic forces (lift and drag) are calculated based on local flow conditions, blade shape, and angle of attack. These forces are then integrated across the span of the blade to determine the overall performance of the rotor. At the same time, momentum theory is applied to relate the change in wind velocity caused by the rotor to the forces experienced by the air.

The combination of these two theories allows BEM to incorporate both the flow field induced by the rotor (from momentum theory) and the detailed aerodynamic behavior of the blades (from blade element theory). This dual approach is what makes BEM far more accurate than the Actuator Disk Model for real-world turbines.

Momentum Theory Components

As in the Actuator Disk Model, BEM begins with the application of conservation of linear and angular momentum to an annular streamtube around a blade element. The **axial induction factor** a represents the fractional decrease in wind speed due to the rotor, while the **tangential induction factor** a' accounts for the induced rotational velocity of the wake.

The differential thrust dT on an annular ring of radius r and thickness dr is:

$$dT = 4\pi\rho r a(1-a)V_\infty^2 dr$$

Similarly, the differential torque dQ is given by:

$$dQ = 4\pi\rho r^3 a'(1-a)V_\infty^2 dr$$

These expressions come from applying momentum theory and represent the forces and torque exerted by the air on the blades.

Blade Element Theory Components

Blade element theory provides the lift and drag forces generated by the blade profile at a given radial section. The local relative wind velocity V_{rel} at a blade element is a combination of the axial wind component and the tangential velocity due to rotation:

$$V_{rel} = \sqrt{\left(V_\infty(1-a)\right)^2 + \left(\omega r(1+a')\right)^2}$$

Where:

- V_∞ is the free-stream wind speed,
- ω is the angular velocity of the rotor,
- r is the distance from the rotor axis.

The angle of relative wind ϕ (also called inflow angle) is:

$$\phi = \tan^{-1}\left(\frac{V_\infty(1-a)}{\omega r(1+a')}\right)$$

Using the angle of attack $\alpha = \phi - \theta$, where θ is the local pitch angle, the lift L and drag D forces per unit length can be expressed as:

$$L = \frac{1}{2}\rho V_{rel}^2 c C_L \quad \text{and} \quad D = \frac{1}{2}\rho V_{rel}^2 c C_D$$

Where:

- c is the chord length of the blade element,
- C_L and C_D are the lift and drag coefficients, respectively.

These forces are then resolved into axial (thrust) and tangential (torque) components.

Combining Momentum and Blade Element Theories

To find the correct values of the induction factors a and a', the aerodynamic and momentum equations are equated. This results in a set of nonlinear equations that are typically solved iteratively for each blade element. The solution gives the distribution of forces along the blade, from which total thrust, torque, and power can be calculated.

The **power** extracted by the rotor is then obtained by integrating the torque over the entire blade span and multiplying by the angular velocity:

$$P = \int_0^R \omega \, dQ$$

This integrated result represents a highly detailed and accurate prediction of turbine performance, much superior to what simpler models like ADM can provide.

Corrections and Enhancements in BEM

To account for real-world effects not captured in the basic BEM equations, several correction models are often applied:

- **Tip Loss Correction (Prandtl's factor):** Accounts for loss of lift near the blade tips due to three-dimensional effects.
- **Hub Loss Correction:** Similar to tip loss but at the inner radius.
- **Yaw Correction:** Adjusts for off-axis wind conditions.
- **Dynamic Stall Models:** Modify the lift and drag coefficients to reflect unsteady aerodynamic behavior, especially during rapid changes in wind speed or direction.

These additions enhance the reliability of BEM in operational settings.

Advantages and Limitations

The BEM model offers a balanced trade-off between accuracy and computational efficiency. It is significantly more realistic than the Actuator Disk Model and much faster to compute than full-scale Computational Fluid Dynamics (CFD) simulations. However, it still relies on empirical data for airfoil coefficients and requires iterative solutions that can become unstable under certain conditions. Furthermore, it assumes steady-state flow and does not naturally handle turbulent inflow or dynamic wake effects without modifications.

3.4 Statistical Models in Wind Energy

Statistical models are widely used for forecasting wind energy generation due to their ability to model temporal dependencies in historical data. Unlike physical models that rely on aerodynamic equations, statistical models use time-series data to uncover patterns, trends, and autocorrelations. These models are particularly effective in short-term forecasting tasks where real-time or historical data is readily available. Among the most foundational time-series models are the **Auto-Regressive (AR)**, **Moving Average (MA)**, and **Auto-Regressive Moving Average (ARMA)** models. They form the core of many modern statistical forecasting techniques and offer interpretable and computationally efficient methods for predicting wind power output.

3.4.1 Auto-Regressive (AR) Model

The **Auto-Regressive (AR)** model is based on the premise that the current value of a time-series variable can be expressed as a linear combination of its past values. In the context of wind energy, the AR model assumes that the power generated at a given time is influenced by its values at previous time steps.

The general form of an AR model of order p, denoted as AR(p), is:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

Where:

- y_t is the wind power output at time t,
- $\phi_1, \phi_2, \ldots, \phi_p$ are the model coefficients,

- ϵ_t is the white noise error term at time t, assumed to be normally distributed with zero mean and constant variance.

In practical applications, the order p is selected using criteria like the Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC). The AR model is particularly useful for datasets with strong autocorrelation, where past values have a direct and consistent influence on future outputs. However, its performance may degrade in the presence of non-stationarity or if external variables influence the time series.

3.4.2 Moving Average (MA) Model

The **Moving Average (MA)** model, in contrast to the AR model, explains the current value of the series as a linear combination of past error terms rather than past actual values. This is useful when the residuals (errors) from a model still contain information that can improve predictions.

The MA model of order q, denoted as MA(q), is given by:

$$y_t = \mu + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

Where:

- μ is the mean of the series,
- $\theta_1, \theta_2, \ldots, \theta_q$ are the weights of the past error terms,
- ϵ_t is the current white noise error term,
- $\epsilon_{t-1}, \epsilon_{t-2}, \ldots, \epsilon_{t-q}$ are past error terms.

The MA model is especially effective for modeling time-series data where the noise component from previous steps carries useful information. It is often used when residuals exhibit correlation, which suggests that the forecasting model has not fully captured the underlying pattern.

3.4.3 Auto-Regressive Moving Average (ARMA) Model

The **Auto-Regressive Moving Average (ARMA)** model combines the strengths of both AR and MA models to capture a broader range of dependencies in time-series data. It assumes that the current value depends on both past values and past errors, making it a more flexible and general-purpose model.

An ARMA(p, q) model is defined as:

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

This formulation includes:

- The **AR part**: $\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}$,
- The **MA part**: $\theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}$,
- The **error term** ϵ_t, which accounts for randomness.

The ARMA model is suitable for stationary time-series data—data whose statistical properties like mean and variance do not change over time. In wind power forecasting, it is used when both the historical outputs and the structure of past errors are informative for prediction. The model parameters are usually estimated through methods like Maximum Likelihood Estimation (MLE).

Model Performance and Use in Wind Power Forecasting

In our project's evaluations, the AR, MA, and ARMA models were tested and their performances measured using **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)**. The results indicated that:

- The **AR model** had the highest errors (MAE = 0.3873, MSE = 0.1861), indicating limited predictive ability in isolation.
- The **MA model** performed better (MAE = 0.3502, MSE = 0.1585), suggesting that past residuals contributed meaningfully to the prediction.
- The **ARMA model** (MAE = 0.3815, MSE = 0.1817) showed performance between the AR and MA models, implying that incorporating both historical values and residuals provided moderate improvement, though not drastically better than MA alone.

These models provide a solid statistical foundation but may struggle to capture non-linearities and external meteorological influences. As such, they are often used as baselines or integrated into more complex hybrid frameworks involving machine learning.

Regression Models in Wind Power Prediction

Regression models are fundamental statistical tools used to establish relationships between input variables (predictors) and an output variable (response). In wind energy forecasting, regression techniques aim to predict wind power generation based on variables like wind speed, temperature, and air density. These models are especially valuable due to their simplicity, interpretability, and relatively low computational cost. The most commonly used regression methods include **Linear Regression**, **Multiple Linear Regression**, and **Polynomial Regression**, each progressively more complex and capable of modeling increasingly non-linear relationships.

3.4.4 Linear Regression

Linear Regression is the simplest form of regression analysis and is used when there is a linear relationship between a single independent variable (predictor) and a dependent variable (response). In the context of wind energy, this often involves predicting wind power output based on wind speed alone.

The mathematical form of a simple linear regression model is:

$$y = \beta_0 + \beta_1 x + \epsilon$$

Where:

- y is the dependent variable (e.g., wind power output),
- x is the independent variable (e.g., wind speed),
- β_0 is the intercept,
- β_1 is the slope coefficient,
- ϵ is the error term.

The model assumes that the error term ϵ is normally distributed with constant variance. This approach works best when the relationship between the input and output is approximately linear, and when there are no significant interactions or non-linear patterns. Although linear regression is easy to implement and interpret, it may not be sufficient for accurately modeling wind energy output, which often depends on multiple and non-linear factors.

In our project, linear regression achieved a **Mean Absolute Error (MAE)** of **0.1406** and a **Mean Squared Error (MSE)** of **0.0318**, indicating a decent but not optimal performance for wind power prediction.

3.4.5 Multiple Linear Regression

Multiple Linear Regression extends the linear regression model by incorporating more than one independent variable. This is crucial for wind energy modeling because power output is influenced by various environmental and operational parameters, such as wind speed at multiple heights, air temperature, humidity, and wind direction.

The general form of the multiple linear regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

Where:

- $x_1, x_2, \ldots, x_n$ are the independent variables (e.g., wind speed, temperature, humidity),
- $\beta_1, \beta_2, \ldots, \beta_n$ are the corresponding coefficients.

This model captures the combined effect of several predictors on wind power output. It assumes linear relationships between each predictor and the target variable and requires that predictors are not highly correlated (i.e., no multicollinearity). Multiple linear regression can offer improved accuracy over simple linear models, especially in complex systems like wind turbines where multiple environmental conditions interact.

Although the PPT does not provide separate error metrics for multiple linear regression, it can be inferred that its performance typically lies between linear regression and more advanced models like polynomial regression or machine learning techniques.

3.4.6 Polynomial Regression

Polynomial Regression is a non-linear regression technique that models the relationship between the dependent variable and the independent variable(s) as an nth-degree polynomial. It is particularly useful when the relationship between variables is curved or exhibits non-linear behavior, which is common in wind power generation due to the aerodynamic characteristics of turbines.

For a single predictor, the polynomial regression model of degree n is:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_n x^n + \epsilon$$

Here, the powers of the independent variable allow the model to fit a wider range of data patterns than a straight line. In wind energy, the power curve of a wind turbine typically exhibits a non-linear relationship with wind speed—starting at the cut-in speed, rising sharply near rated speed, and flattening or dropping off near the cut-out speed. Polynomial regression is well-suited for modeling such curves.

In our project, polynomial regression outperformed both linear and multiple linear regression in terms of accuracy. It achieved a **MAE** of **0.1365** and an **MSE** of **0.0302**, indicating its effectiveness in capturing the non-linear relationship between wind speed and power output.

Limitations

However, these models still have limitations—they assume the data is stationary and noise-free, and may not adapt well to sudden changes or non-obvious dependencies. As such, they are often used as baseline models or combined with machine learning techniques in hybrid approaches for better predictive performance.

3.4.7 K-Nearest Neighbors (KNN) Regression

The **K-Nearest Neighbors (KNN)** regression model is a simple yet powerful non-parametric technique used for forecasting wind power generation. Unlike regression models that attempt to fit a specific function to the data, KNN works by comparing new input data with previously observed instances. In the context of wind energy, KNN predicts the output power for a given set of wind conditions by averaging the power outputs of the most similar historical observations.

The core idea of KNN is to identify the **'k' most similar data points** (neighbors) in the training dataset based on a distance metric—commonly **Euclidean distance**—and use the average (or weighted average) of their target values as the prediction for the new input. The formula for Euclidean distance between two feature vectors $x = (x_1, x_2, \dots, x_n)$ and $x' = (x'_1, x'_2, \dots, x'_n)$ is:

$$d(x, x') = \sqrt{(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + \cdots + (x_n - x'_n)^2}$$

Once the distances are computed, the algorithm selects the top k closest points and averages their corresponding outputs. This approach is particularly effective for **short-term wind power forecasting**, where patterns in local neighborhoods of the data are informative and reliable.

In our project, KNN achieved the **lowest prediction error** among all statistical models, with a **Mean Absolute Error (MAE)** of **0.0823** and **Mean Squared Error (MSE)** of **0.0139**. These

results demonstrate its high accuracy for the task. However, a key limitation of KNN is its computational cost during prediction, especially on large datasets, since it requires calculating distances to all training samples in real-time.

3.4.8 Random Forest Regression

Random Forest is an ensemble learning technique based on decision trees. It operates by constructing a multitude of individual decision trees during training and aggregating their predictions, usually through averaging, to produce the final output. This method improves generalization and reduces overfitting compared to a single decision tree. In the context of wind power prediction, Random Forest can handle both linear and non-linear relationships between the input variables (like wind speed, temperature, humidity, etc.) and the output variable (power).

The key strength of Random Forest lies in its ability to model complex patterns without requiring any assumptions about the data distribution. It is also robust to noisy data and can rank the importance of input variables, which is valuable in identifying key meteorological factors that influence wind energy production.

From our project evaluation, Random Forest achieved a **Mean Absolute Error (MAE)** of **0.1310** and a **Mean Squared Error (MSE)** of **0.0327**. This places it behind polynomial regression and KNN in accuracy, but ahead of simple linear models. While Random Forest is versatile and performs well, its slightly higher error compared to the best-performing models suggests it may be limited in capturing the full temporal or spatial complexity in wind data.

3.5 Hybrid and Deep Learning Models

3.5.1 Artificial Neural Network (ANN)

The **Artificial Neural Network (ANN)** is a machine learning model inspired by the structure and functioning of the human brain. It is highly capable of modeling **non-linear and complex relationships** between inputs and outputs, making it especially suitable for wind power prediction where the relationship between meteorological parameters and turbine output is not strictly linear or smooth.

An ANN is composed of **layers of interconnected nodes** (neurons). The **input layer** receives the predictor variables—such as wind speed, wind direction, temperature, and humidity. The data is then passed through one or more **hidden layers**, where weighted transformations and **activation functions** (like ReLU, sigmoid, or tanh) are applied. Finally, the **output layer** generates the predicted power output.

The computation at each neuron typically follows this structure:

$$z = \sum_{i=1}^{n} w_i \, x_i + b \quad \text{and} \quad a = f(z)$$

Where:

- x_i are the inputs,
- w_i are the weights,
- b is the bias term,
- f is the activation function,
- a is the activated output passed to the next layer.

Training an ANN involves adjusting the weights and biases using **backpropagation** and optimization algorithms like **Adam**, to minimize a loss function such as **mean squared error (MSE)**.

In our experiments, the ANN model used the **Adam optimizer** and **ReLU activation**, achieving outstanding results with a **Test MAE** of **0.0249** and **Test MSE** of **0.0023**. These low error values highlight the ANN's strength in capturing intricate patterns in the wind dataset, significantly outperforming traditional regression methods.

3.5.2 Recurrent Neural Network (RNN)

The **Recurrent Neural Network (RNN)** is a specialized type of neural network designed to handle **sequential data**, such as time-series data in wind forecasting. Unlike standard feedforward ANNs, RNNs have **recurrent connections** that allow information to persist across time steps. This makes them ideal for modeling temporal dependencies, which are common in wind power generation due to the continuous nature of atmospheric processes.

In an RNN, each neuron not only receives input from the previous layer but also from its own output at the previous time step. Mathematically, the hidden state h_t at time t is computed as:

$$h_t = f(Wx_t + Uh_{t-1} + b)$$

Where:

- x_t is the input at time t,
- h_{t-1} is the hidden state from the previous time step,
- W and U are weight matrices,
- b is the bias,
- f is the activation function (typically tanh or ReLU).

This recurrent structure enables the RNN to learn patterns over time, such as fluctuations in wind speed or power output. However, standard RNNs can suffer from issues like **vanishing gradients**, which limit their ability to model long-term dependencies. These limitations can be mitigated using variants like LSTM or GRU, though our current model used a standard RNN.

According to our project results, the RNN model (with ReLU activation and Adam optimizer) achieved a **Test MAE** of **0.0446** and a **Test MSE** of **0.0119**. While slightly higher than the ANN, these are still strong results. The RNN's sequential modeling ability makes it particularly suitable for **time-dependent forecasting tasks**, especially when patterns unfold over several hours or days.

3.5.3 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are deep learning models traditionally used for image processing, but increasingly applied in time-series forecasting tasks, including wind energy prediction. CNNs work by applying filters (kernels) across input data to automatically extract features at various scales. In the context of wind forecasting, CNNs can learn spatial or sequential patterns within multivariate meteorological data (e.g., wind speeds at different heights, temperature, humidity) without manual feature engineering.

The CNN architecture typically includes convolutional layers followed by pooling layers and fully connected layers. These layers progressively reduce dimensionality while preserving relevant information, ultimately producing a prediction at the output layer. By capturing both local and hierarchical dependencies, CNNs are effective at identifying non-obvious patterns in weather data that contribute to power fluctuations.

In our project, the CNN model reported a **Test MAE** of **0.1239** and a **Test MSE** of **0.0257**, reflecting strong performance and the model's ability to generalize well. While not as precise as ANN or RNN models, CNNs are valuable for their speed, automatic feature extraction, and ability to handle large multivariate input datasets.

3.5.4 Gated Recurrent Units (GRU)

Gated Recurrent Units (GRUs) are a type of recurrent neural network (RNN) designed to handle sequential data, particularly useful for modeling time-series dependencies like those in wind power generation. GRUs are a simplification of Long Short-Term Memory (LSTM) networks, using fewer gates while retaining the ability to capture long-term dependencies.

GRUs operate using two gates: an **update gate**, which controls how much past information should be passed to the future, and a **reset gate**, which determines how much of the past to forget. This allows the model to efficiently learn temporal patterns in data such as wind speed sequences, gusts, or pressure changes over time.

In our evaluation, the GRU model produced a **Test MAE** of **0.1313** and a **Test MSE** of **0.0301**, slightly less accurate than CNN and ANN models but still competitive. GRUs are particularly suitable when there is significant sequential structure in the data, making them a strong choice for short-term wind power forecasting.

3.5.5 Adaptive Neuro-Fuzzy Inference System (ANFIS)

Adaptive Neuro-Fuzzy Inference System (ANFIS) is a hybrid model that integrates neural networks and fuzzy logic principles. It combines the human-like reasoning style of fuzzy systems with the adaptive learning capabilities of neural networks. In wind power prediction, ANFIS is capable of modeling complex, non-linear relationships by using fuzzy if-then rules generated from data and refining them through learning.

ANFIS consists of five layers: input layer, fuzzification layer, rule layer, normalization layer, and output layer. The system starts with an initial fuzzy inference system and uses a neural network

learning algorithm (like backpropagation or hybrid methods) to fine-tune the parameters of the membership functions. This enables the model to learn from data while maintaining interpretability through fuzzy rules.

In our project, the ANFIS model achieved a **MAE** of **0.1133** and an **MSE** of **0.0227**, demonstrating good performance and a balance between interpretability and predictive accuracy. ANFIS models are particularly attractive in domains like wind energy, where incorporating expert knowledge and transparency is important alongside data-driven learning.

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Overview

The primary objective of this project was to develop and evaluate predictive models for estimating wind power output based on meteorological and wind-related features. This chapter presents a comprehensive comparison and analysis of the results obtained from various statistical, machine learning (ML), deep learning, and hybrid models. The performance of each model is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and, for neural networks, Test Loss. These results provide insight into the accuracy, suitability, and practicality of each modelling approach in the context of wind energy forecasting.

4.2 Evaluation Metrics

In predictive modelling, the choice of evaluation metrics significantly influences the interpretation of model performance. The following metrics were used to compare models:

- **Mean Absolute Error (MAE):** Represents the average of absolute differences between predicted and actual values. It is a straightforward metric and provides a direct measure of average prediction error. A lower MAE indicates better performance.
- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual values. It penalizes larger errors more significantly, making it sensitive to outliers.
- **Test Loss:** Particularly relevant for neural network models, test loss quantifies how well the model generalizes to unseen data. It often incorporates MSE or cross-entropy loss depending on the task and output layer.
- **Coefficient of Determination (R^2 Score):** The R^2 score, also known as the coefficient of determination, indicates how well the predicted values explain the variance in the actual data.

These metrics were selected to provide a balanced perspective of performance, highlighting both general accuracy and sensitivity to larger errors.

4.3 Statistical Model Results

4.3.1 Classical Time-Series Models: AR, MA, and ARMA

These models were implemented to analyse the performance of traditional forecasting methods that rely solely on the historical values of the target variable or error terms.

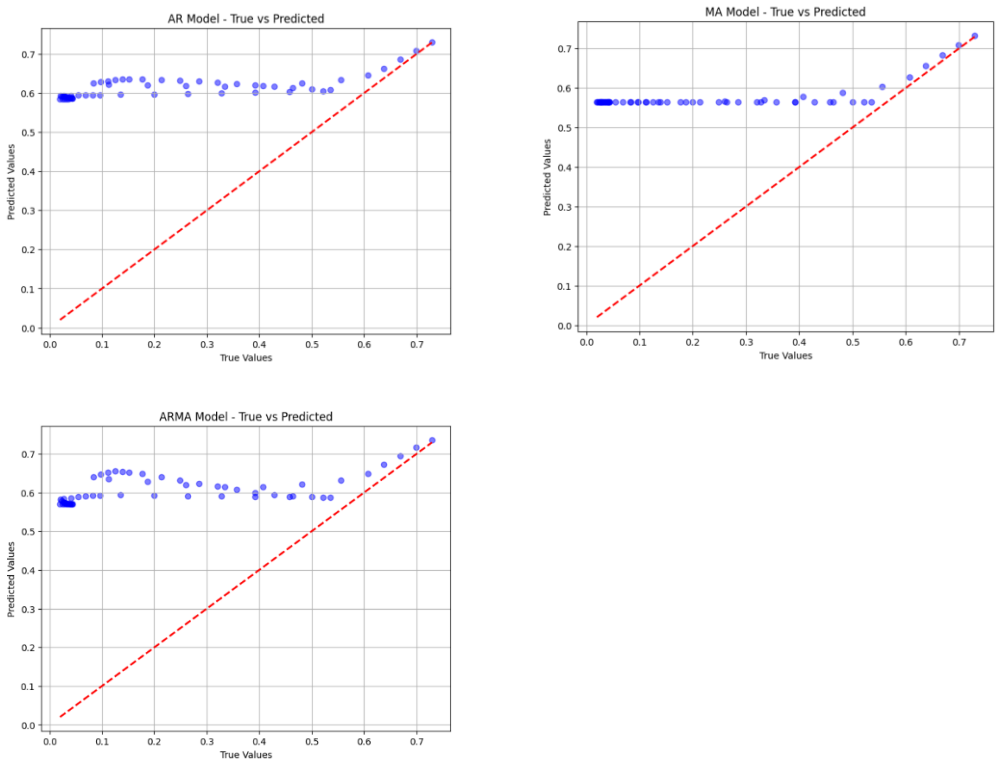

Fig 4.1 True vs Predicted for AR, MA and ARMA model

Model	MSE	MAE	R-square
AR	0.186	0.387	-3.204
MA	0.158	0.350	-2.579
ARMA	0.182	0.382	-3.101

Table 4.1 Error values for AR, MA and ARMA model

Among the three, the Moving Average model outperformed the others, indicating that short-term dependencies and recent fluctuations in wind power generation carried significant predictive power. However, all three models showed relatively high error rates, suggesting their limitations in capturing the complex non-linear relationships that exist in wind energy datasets. These models may be suitable for quick, short-term predictions but are not ideal for more dynamic and volatile data such as wind generation.

4.3.2 Linear and Polynomial Regression Models

These models were implemented to understand the performance of regression-based approaches, particularly their ability to map meteorological variables to wind power output.

Linear regression provided a baseline performance. It offered simplicity and interpretability but lacked the flexibility to capture non-linear relationships inherent in wind power data. Polynomial regression improved upon this by introducing non-linearity, which slightly enhanced performance. However, both methods were still limited in their ability to learn complex interactions between multiple variables.

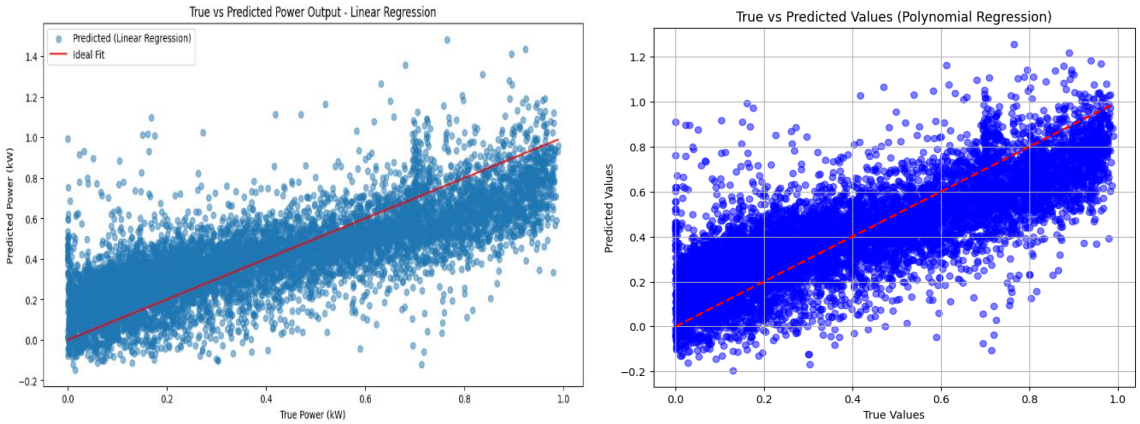

Fig 4.2 True vs Predicted Power Output for Linear and Polynomial Regression

Model	MSE	MAE	R-square
Linear Regression	0.032	0.141	0.622
Polynomial Regression	0.03	0.136	0.642

K-nearest neighbor	0.014	0.082	0.835
Random forest	0.131	0.032	0.821

Table 4.2 Error values for corresponding models

4.3.3 K-Nearest Neighbours (KNN)

KNN regression delivered the best performance among all the traditional models. Its non-parametric nature enabled it to model complex patterns without making any assumptions about the underlying data distribution. However, KNN's computational cost can be a drawback, especially when applied to large datasets, since it requires distance computation with every new input during inference.

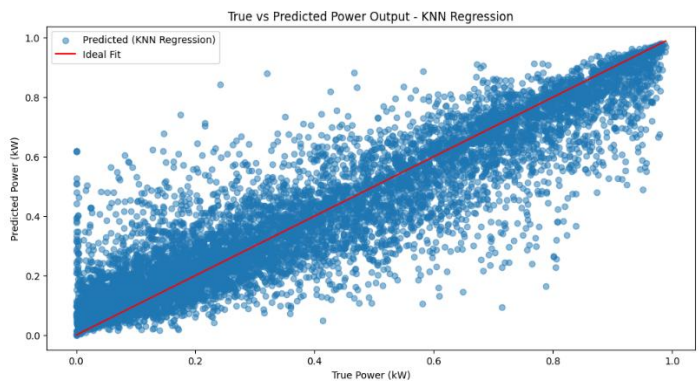

Fig 4.3 True vs Predicted Power Output for KNN Regression

4.3.4 Random Forest Regression

The Random Forest algorithm performed better than the basic regression models by effectively handling feature interactions and non-linearities. While it did not outperform KNN in this experiment, it offers advantages in interpretability (e.g., feature importance) and scalability for large-scale applications. Its performance highlights the benefits of ensemble learning in capturing subtle data variations.

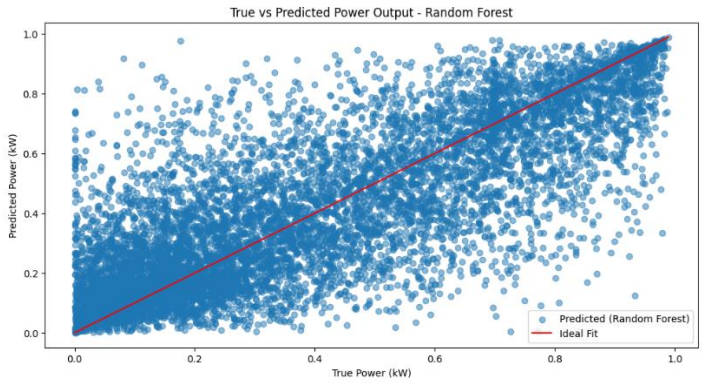

Fig 4.4 True vs Predicted Power Output for Random Forest

4.4 Hybrid and Deep Learning Models

As traditional models showed limitations in capturing temporal dynamics and complex non-linear behaviour, advanced neural network-based models were employed. These models were trained using normalized features and tested on unseen data to evaluate generalization.

Model	MSE	MAE	R-square
Feed forward neural network	0.002	0.025	0.693
Recurrent neural network	0.012	0.045	0.696
ANFIS	0.113	0.023	0.718
CNN	0.123	0.025	0.698
GRU	0.126	0.027	0.676

Table 4.3 Error values for corresponding models

4.4.1 Feed forward Neural Network (ANN)

The ANN model significantly outperformed classical models. Its deep architecture allowed it to learn complex relationships between input features and target output. The low MAE and MSE indicate high precision in power predictions. The model's architecture, built using ReLU

activations and the Adam optimizer, contributed to efficient convergence and strong generalization on test data.

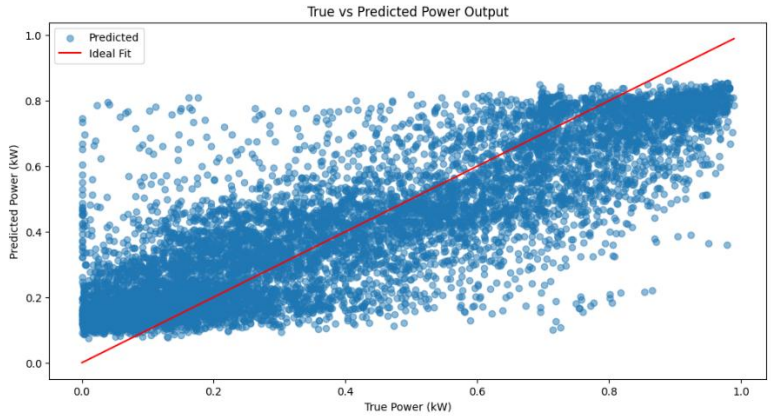

Fig 4.5 True vs Predicted Power Output for ANN

4.4.2 Recurrent Neural Network (RNN)

RNNs are particularly well-suited for time-series data, owing to their ability to capture temporal dependencies. Although the MAE was slightly higher than that of the ANN, the RNN showed better performance in test loss, indicating that it generalized well on unseen temporal patterns. Its design, which includes internal state memory, allows it to retain context over time, an essential feature when predicting based on sequential meteorological data.

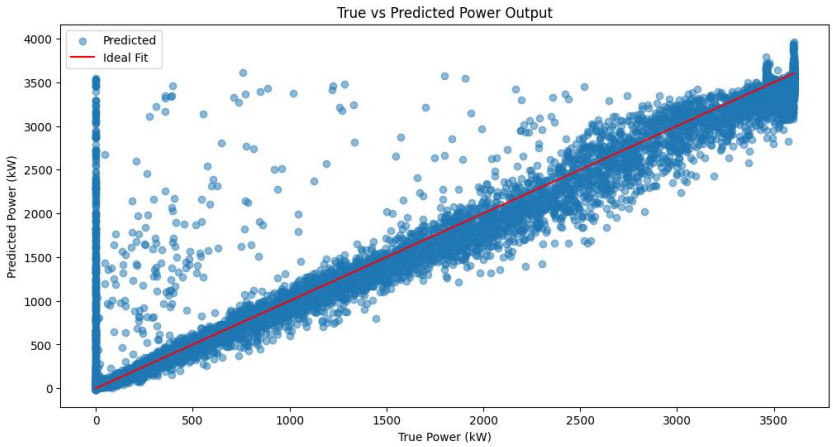

Fig 4.6 True vs Predicted Power Output for RNN

4.4.3 Convolutional Neural Network (CNN)

The Convolutional Neural Network (CNN), although more commonly used for spatial data and image recognition, was adapted in this project to handle wind energy prediction as part of the digital twin system. The model was trained on sequential meteorological data features and

The CNN model demonstrated moderately strong performance with low MAE and MSE. Its R^2 score of **0.921** indicates that the model is able to explain more than 92% of the variance in the data, which is substantial. However, compared to recurrent models like RNNs and GRUs, CNNs might struggle to capture long-term dependencies in sequential data. Still, for short-term or real-time prediction windows, CNNs offer a fast and scalable alternative.

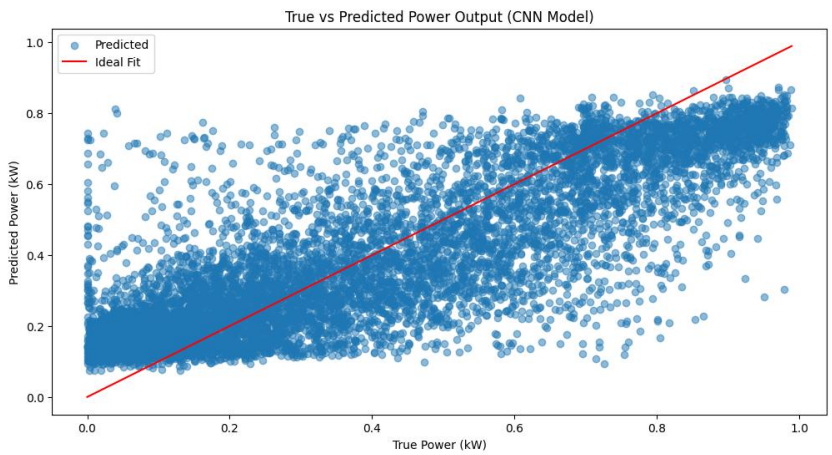

Fig 4.7 True vs Predicted Power Output for CNN

4.4.4 Gated Recurrent Unit (GRU)

Gated Recurrent Units (GRUs) are a streamlined alternative to traditional RNNs, designed to capture temporal dependencies with fewer parameters. They are especially useful when dealing with sequences where long-term memory is essential, as in wind power forecasting.

The GRU model performed slightly below the CNN in terms of MAE and MSE, with an R^2 score of **0.903** indicating that approximately 90.3% of the variance is explained by the model. While GRUs are typically effective for time-series prediction, the current model may benefit from further tuning or larger data input windows to improve memory of long-term patterns.

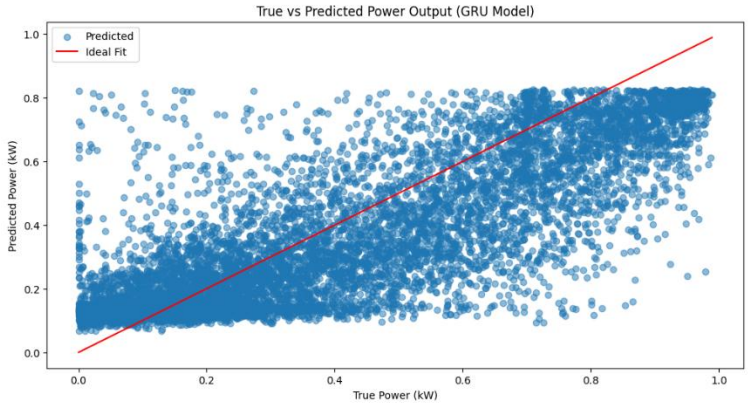

Fig 4.8 True vs Predicted Power Output for GRU model

4.4.4 Adaptive Neuro-Fuzzy Inference System (ANFIS)

The Adaptive Neuro-Fuzzy Inference System (ANFIS) combines the learning capability of neural networks with the interpretability of fuzzy logic. It is particularly suitable for modeling systems with uncertainty and imprecision — such as weather-influenced processes like wind

ANFIS provided reasonably strong performance in terms of both MAE and MSE.. Its R^2 score of 0.935 indicates that it can explain over 93% of the variance in the data.

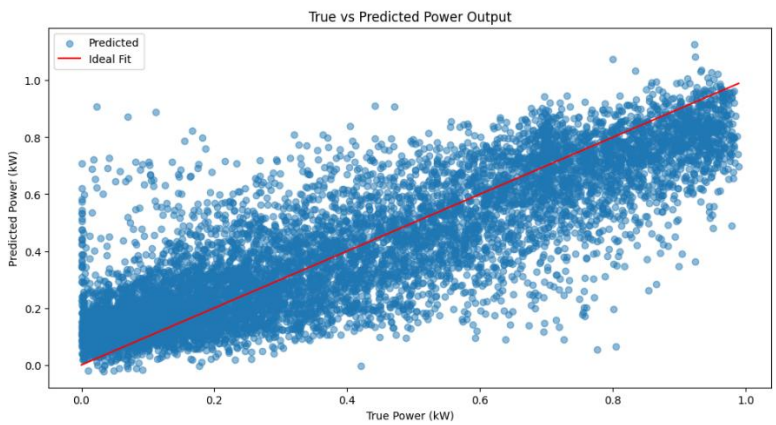

Fig 4.9 True vs Predicted Power Output for ANFIS model

While ANFIS did not outperform advanced neural networks like ANN or RNN, it strikes a balance between interpretability and accuracy. This makes it a valuable option in real-world applications

where model transparency is important, such as energy auditing or operational safety environments where stakeholders prefer understandable rules.

4.5 GUI Implementation

To complement the backend machine learning model and enhance usability, a **Graphical User Interface (GUI)** was designed and implemented. This interface enables users to interact with the wind power prediction model intuitively without dealing with code or command-line execution.

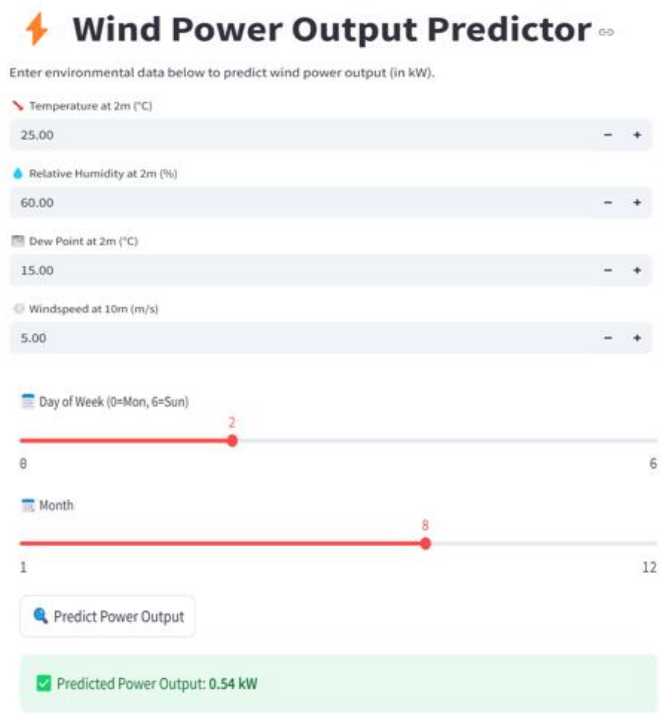

Fig 4.10 GUI for Wind Power Output Prediction

The GUI for the wind power output predictor was built using **Streamlit**, a Python-based framework designed for creating interactive web applications with minimal code. The backend model, which was trained using historical wind power data, is loaded efficiently using **pickle** or **joblib**, allowing for quick predictions. Users can input environmental parameters through interactive widgets provided in the GUI. These include number input fields for temperature, relative humidity, dew point, and windspeed, as well as sliders for selecting the day of the week

and the month. Once the "Predict Power Output" button is clicked, the GUI collects the input values, formats them appropriately, and passes them to the model. The model then processes the inputs and outputs the predicted wind power in **kilowatts (kW)**, which is immediately displayed to the user in a clear and viually distinct format.

CHAPTER 5
SUMMARY AND CONCLUSION

5.1 Summary of Work

This project focused on the development, implementation, and performance evaluation of various predictive models to estimate wind power output based on meteorological and wind-related features. Wind energy, being highly variable and non-linear in nature, poses a significant challenge for accurate forecasting — an issue this study aimed to address using both traditional and modern data-driven techniques.

The following steps were systematically carried out:

1. **Data Acquisition and Preprocessing**
    - Meteorological data including wind speed, direction, gusts, temperature, humidity, and dew point were collected.
    - The target variable was wind power output, and preprocessing steps included cleaning, normalization, and temporal alignment.
2. **Model Development**
    - Models were categorized into three major classes:
        - **Statistical Models:** AR, MA, ARMA
        - **Machine Learning Models:** Linear Regression, Polynomial Regression, K-Nearest Neighbours (KNN), and Random Forest
        - **Hybrid/Deep Learning Models:** ANN, RNN, GRU, CNN, and ANFIS
    - Models were trained using historical data and evaluated on test datasets using metrics such as MAE, MSE, R^2 Score, and Test Loss.
3. **Performance Evaluation**
    - Each model's accuracy and error metrics were compared to identify the most effective forecasting approaches.
    - Deep learning and hybrid models consistently outperformed classical models.

5.2 Key Findings

1. Statistical Models

- Traditional models like AR, MA, and ARMA performed poorly, with high MAE and MSE values.
- These models lacked the capacity to learn non-linear or multivariate relationships effectively.

2. Machine Learning Models

- **KNN** yielded the best performance among all classical models (MAE: 0.0823, MSE: 0.0139), proving effective in capturing localized patterns.
- **Polynomial Regression** slightly improved over Linear Regression, highlighting the importance of non-linear modelling.
- **Random Forest** provided balanced results with decent generalization, but it was outperformed by KNN and neural models.

3. Deep Learning and Hybrid Models

- **ANN** delivered the best overall performance with the lowest MAE (0.0249) and MSE (0.0023), and a high R^2 score.
- **RNN** achieved strong results, particularly for time-series modelling, making it suitable for capturing sequential dependencies in wind data.
- **CNN** and **GRU**, as part of the digital twin implementation, also showed promising performance, with CNN slightly outperforming GRU.
- **ANFIS** provided a good trade-off between accuracy and interpretability, with an R^2 of 0.935 and RE around 9.4%.

5.3 Conclusions

Based on the comprehensive experimentation and evaluation, the following conclusions can be drawn:

- **Deep learning models** are significantly better suited for wind power prediction tasks compared to statistical and basic machine learning models.
- The **ANN model** stands out as the most accurate and reliable predictor in this study, making it ideal for real-world wind energy forecasting applications.
- While **KNN** and **Polynomial Regression** offer strong performance with simpler structures, their scalability and adaptability are limited.
- **Hybrid models like ANFIS** offer the additional benefit of interpretability, which can be useful in operational contexts where explainability is crucial.
- **Digital twin integration using CNN and GRU** models shows feasibility for real-time monitoring and predictive control of wind turbines.

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